Homework 4

MATH 441: DIFFERENTIAL EQUATIONS

NAME: DATE:

(Exercises are taken from *Elementary Differential Equations and Boundary Value Problems* by William E. Boyce and Richard C. DiPrima)

In each of Problems 1 and 2, find the solution of the given initial value problem. Sketch the graph of the solution and describe its behavior as t increases.

1. Exercise §3.1 #11.

$$6y'' - 5y' + y = 0, \qquad y(0) = 4, \qquad y'(0) = 0$$

2. Exercise §3.1 #12.

$$y'' + 3y' = 0, \qquad y(0) = -2, \qquad y'(0) = 3$$

In each of Problems 3 and 4, determine the linges interval in which the given intial value problem iscertain to have a unique twice-differentiable solution. Do not attempt to find the solution. Explain your answer.

3. Exercise §3.2 #10.

$$y'' + (\cos t)y' + 3(\ln|t|)y = 0, \qquad y(2) = 3, \qquad y'(2) = 1$$

4. Exercise §3.2 #11.

$$(x-3)y'' + xy' + (\ln|x|)y = 0, \qquad y(1) = 0, \qquad y'(1) = 1$$

5. Exercise §3.3 #19. Find the solution of the given initial value problem. Sketch the graph of the solution and describe its behavior for increasing t

$$y'' - 2y' + 5y = 0, \qquad y(\pi/2) = 0, \qquad y'(\pi/2) = 2$$

6. Exercise §3.4 #15(a). Solve the initial value problem and plot its solution for $0 \leq t \leq 5$.

$$4y'' + 12y' + 9y = 0, \qquad y(0) = 1, \qquad y'(0) = -4$$